

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

The function $f(x)$ is defined as **19** more than **4** times a number x . If $y = f(x)$ is graphed in the xy -plane, what is the best interpretation of the x -intercept?

Answer

- A. When $f(x) = 0$, the number is $-\frac{19}{4}$.
- B. When the number is **0**, $f(x) = 19$.
- C. The value of $f(x)$ increases by **1** for each increase of **4** in the value of the number.
- D. For each increase of **1** in the value of the number, $f(x)$ increases by **4**.

Correct Answer: A

Rationale

Choice A is correct. It’s given that the function $f(x)$ is defined as **19** more than **4** times a number x . This can be represented by the equation $f(x) = 4x + 19$. The x-intercept of the graph of $y = f(x)$ in the xy -plane is the point where the graph intersects the x -axis, or the point on the graph where the value of $f(x)$ is equal to **0**. Substituting **0** for $f(x)$ in the equation $f(x) = 4x + 19$ yields $0 = 4x + 19$. Subtracting **19** from each side of this equation yields $-19 = 4x$. Dividing each side of this equation by **4** yields $x = -\frac{19}{4}$. Therefore, when $f(x) = 0$, the number is $-\frac{19}{4}$.

Choice B is incorrect. This is the best interpretation of the y-intercept, not the x-intercept, of the graph of the function.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the best interpretation of the slope, not the x-intercept, of the graph of the function.